

DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis

Birgit Pfitzmann
IBM Research
Rueschlikon, Switzerland
bpf@zurich.ibm.com

Christoph Auer
IBM Research
Rueschlikon, Switzerland
cau@zurich.ibm.com

Michele Dolfi
IBM Research
Rueschlikon, Switzerland
dol@zurich.ibm.com

Ahmed S. Nassar
IBM Research
Rueschlikon, Switzerland
ahn@zurich.ibm.com

Peter Staar
IBM Research
Rueschlikon, Switzerland
taa@zurich.ibm.com

ABSTRACT

Accurate document layout analysis is a key requirement for high-quality PDF document conversion. With the recent availability of public, large ground-truth datasets such as PubLayNet and DocBank, deep-learning models have proven to be very effective at layout detection and segmentation. While these datasets are of adequate size to train such models, they severely lack in layout variability since they are sourced from scientific article repositories such as PubMed and arXiv only. Consequently, the accuracy of the layout segmentation drops significantly when these models are applied on more challenging and diverse layouts. In this paper, we present *DocLayNet*, a new, publicly available, document-layout annotation dataset in COCO format. It contains 80863 manually annotated pages from diverse data sources to represent a wide variability in layouts. For each PDF page, the layout annotations provide labelled bounding-boxes with a choice of 11 distinct classes. *DocLayNet* also provides a subset of double- and triple-annotated pages to determine the inter-annotator agreement. In multiple experiments, we provide baseline accuracy scores (in mAP) for a set of popular object detection models. We also demonstrate that these models fall approximately 10% behind the inter-annotator agreement. Furthermore, we provide evidence that *DocLayNet* is of sufficient size. Lastly, we compare models trained on PubLayNet, DocBank and *DocLayNet*, showing that layout predictions of the *DocLayNet*-trained models are more robust and thus the preferred choice for general-purpose document-layout analysis.

CCS CONCEPTS

• **Information systems** → **Document structure**; • **Applied computing** → **Document analysis**; • **Computing methodologies** → **Machine learning**; **Computer vision**; **Object detection**;

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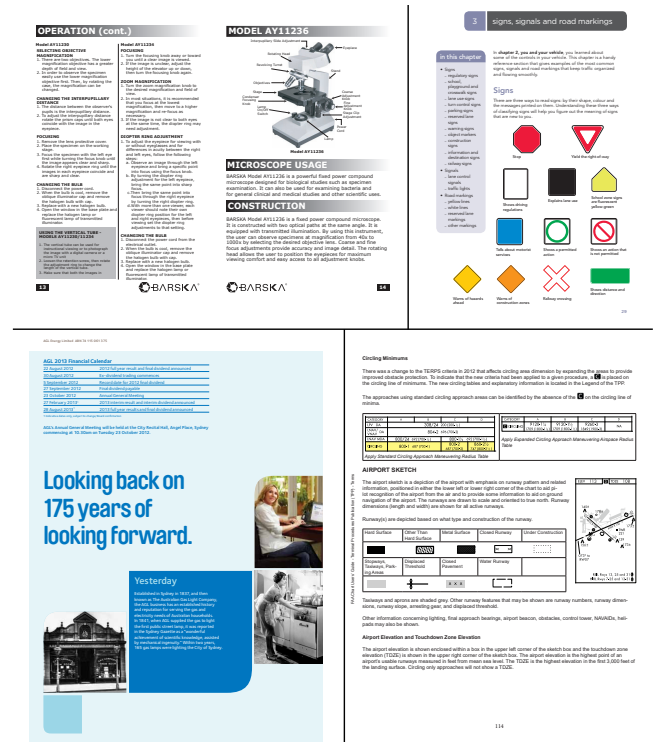


Figure 1: Four examples of complex page layouts across different document categories

KEYWORDS

PDF document conversion, layout segmentation, object-detection, data set, Machine Learning

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